ChatGPT

Universal Deep Research Framework

Executive summary

When the topic is not yet specified, the strongest default is not to jump straight into conclusions, but to use a staged evidence process: begin with a **scoping pass** to map concepts, definitions, source types, and research gaps; convert that into a **protocol-driven focused review** once the question is refined; then conduct **structured appraisal, extraction, synthesis, and confidence grading** before writing conclusions. That sequence reflects mainstream evidence-synthesis practice across systematic reviews, scoping reviews, and review standards bodies. PRISMA 2020 is the reporting baseline; PRISMA-ScR supports broad mapping exercises; Cochrane, Campbell, and JBI provide conduct guidance; and PRISMA-S and PRESS strengthen search transparency and reproducibility. [1]

For an unspecified topic, the most useful deliverable is therefore a **reusable research blueprint**: clear objectives, scoping logic, question and hypothesis templates, search methods, source priorities, screening rules, extraction fields, synthesis choices, confidence ratings, a timeline, and a standard output package. This report provides exactly that, with worked examples that can be adapted to science, engineering, policy, business, health, social science, or mixed-method topics. [2]

The core principle throughout is simple: **prioritize primary and official sources whenever they exist**, use broad indexing tools to avoid blind spots, document searches exactly as run, assess risk of bias with tools matched to study design, and separate **what the evidence says** from **how confident you should be in it**. That combination is what makes a deep research report reproducible rather than merely well written. [3]

Research objectives and scope

A strong deep-research project begins by deciding **what kind of question** is being asked. If the goal is to estimate effect, compare interventions, or test a causal hypothesis, the review should be structured around an effect question. If the goal is to map a field, clarify definitions, or find research gaps, a scoping review is usually the better first move. PRISMA-ScR explicitly frames scoping reviews as a way to synthesize evidence, assess the extent of the literature, and determine whether a full systematic review is warranted. Cochrane and Campbell, by contrast, are oriented toward questions that need formal appraisal and synthesis of effect evidence. [4]

For focused effectiveness questions, a practical default is **PICO**: Population, Intervention, Comparator, Outcome. Cochrane's own PICO search guidance is built around these components. For broad evidence-mapping questions, a common scoping structure is **PCC**: Population, Concept, Context. That structure is widely used in JBI-aligned scoping review practice and in PRISMA-ScR reporting. [5]

A universal objective template looks like this:

Objective type	Best when you need to know	Typical framing
Effectiveness or causal	Whether X changes Y	“Among **[population]**, does **[intervention/ exposure]** compared with **[comparator]** affect **[outcome]**?”
Descriptive or prevalence	How common or widespread something is	“What is the prevalence, distribution, or adoption of **[phenomenon]** in **[setting]**?”
Mechanism or explanatory	Why or how something happens	“Through what mechanisms does **[factor]** influence **[outcome]**?”
Scoping or landscape	What evidence exists and where gaps remain	“What research exists on **[concept]** in **[context]**, and what patterns or gaps emerge?”
Policy or regulatory	What rules, standards, or guidance currently control the space	“What current laws, standards, and regulatory guidance govern **[topic]**?”

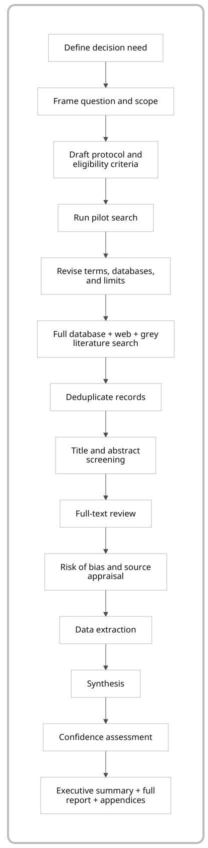
A good hypothesis template should also be specified **before** most searching and certainly before synthesis. Examples:

- **Effect hypothesis:** “Intervention X improves Outcome Y relative to Comparator Z.”
- **Null hypothesis:** “There is no meaningful difference in Outcome Y between X and Z.”
- **Risk hypothesis:** “Exposure X is associated with a higher likelihood of Outcome Y.”
- **Implementation hypothesis:** “Barrier A and facilitator B explain variation in adoption across settings.”
- **Market or policy hypothesis:** “Recent regulatory and cost changes have shifted adoption toward X.”

The scope statement should decide, at minimum, topic boundaries, time range, study types, geographies, languages, outcomes of interest, and what counts as “decision-grade” evidence. That scope should be written down as a protocol, because PRISMA-P was developed specifically to improve the transparency and completeness of systematic review protocols, while OSF and PROSPERO support time-stamped preregistration and reduce duplication. PROSPERO describes itself as the first prospective register of systematic reviews in health and social care, and OSF registrations create a time-stamped, read-only record. [6]

Methodology and search design

A generic deep-research workflow should be explicit, auditable, and reproducible. PRISMA 2020 emphasizes transparent reporting of why the review was done, what was done, and what was found; PRISMA-S adds a dedicated 16-item checklist for reporting literature searches; and PRESS provides an evidence-based checklist for peer-reviewing electronic search strategies. Cochrane complements these by detailing review conduct, including searching, selection, data collection, synthesis preparation, and meta-analysis. [7]



The workflow above is aligned with Cochrane review conduct, PRISMA-style reporting, and PRISMA-S search documentation. [8]

A default search design should combine:

1. **Structured database searching** for peer-reviewed literature.
2. **Targeted official-site searching** for laws, guidance, standards, statistics, and institutional documents.
3. **Grey literature searching** for reports, white papers, technical briefs, theses, working papers, and implementation documents.
4. **Registry and citation searching** where topic-appropriate.
5. **Backward and forward citation chasing** for completeness. [9]

A practical set of inclusion and exclusion criteria for an unspecified topic can be written in reusable form:

Criterion group	Include	Exclude
Relevance	Directly addresses the research question or a prespecified subquestion	Only tangential or metaphorical mention of the topic

Criterion group	Include	Exclude
Source credibility	Peer-reviewed studies, official institutional documents, validated datasets, standards, patents, trial registries, reputable industry reports	Anonymous blogs, unsourced opinion pieces, promotional pages without underlying evidence
Time window	Default: last 5–10 years for fast-moving topics; longer for foundational theory and historical context	Superseded material unless historically important
Language	English by default; expand when critical evidence is likely in other languages	Languages the team cannot translate or verify, unless translation is feasible
Evidence type	Predefined study designs and document types relevant to the objective	Designs incapable of answering the question as framed
Quality threshold	Studies or reports with transparent methods, identifiable authorship, and inspectable data or citations	Opaque methods, irreproducible claims, or obvious conflicts without disclosure

A rigorous search should be piloted first, then refined. PRESS specifically checks whether the search strategy correctly translates the question, uses Boolean and proximity operators appropriately, applies controlled vocabulary and text words well, and documents limits and sources. PRISMA-S further expects the databases, online resources, registries, full strategies, limits, peer review, and deduplication steps to be reported. [10]

Suggested academic search strings

Use these as templates, then replace bracketed placeholders.

General multidisciplinary template

```
("[topic]" OR [synonym1] OR [synonym2] OR [broader term])
AND
("[outcome]" OR [measure] OR [effect] OR [indicator])
AND
([population] OR [setting] OR [sector])
NOT
([known irrelevant meaning] OR [homonym])
```

Intervention or effect template

```
([population])
AND
```

```
([intervention/exposure] OR [synonyms])
AND
([comparator] OR usual care OR baseline OR control)
AND
([outcome1] OR [outcome2] OR [indicator*])
```

Scoping or landscape template

```
([concept] OR [related concept] OR [adjacent term])
AND
([context] OR [sector] OR [jurisdiction] OR [application])
AND
(review OR evidence OR study OR trial OR implementation OR policy)
```

Engineering and computing template

```
([technology] OR [system] OR [architecture])
AND
(performance OR reliability OR safety OR usability OR cost OR scalability)
AND
(experiment* OR benchmark* OR evaluation OR case study OR field trial)
```

Illustrative example

```
("large language model*" OR chatbot* OR "AI tutor*" OR "artificial
intelligence")
AND
("higher education" OR universit* OR college OR undergraduate*)
AND
("learning outcome*" OR achievement OR grade* OR engagement OR retention)
NOT
(marketing OR "customer service")
```

Suggested web and official-site queries

For web search, exact phrase matching and domain restriction are often more effective than long Boolean chains.

Official guidance and regulation

```
site:gov OR site:regulator-domain "[topic]" guidance filetype:pdf
site:who.int "[topic]" report
site:fda.gov "[topic]" guidance
```

Standards and patents

```
site:iso.org "[topic]" standard
site:wipo.int PATENTSCOPE "[topic]"
site:uspto.gov patent "[topic]"
```

Policy and statistics

```
site:worldbank.org "[topic]" data
site:oecd.org "[topic]" report
site:ministry-domain "[topic]" policy pdf
```

Grey literature and implementation evidence

```
"[topic]" report filetype:pdf
"[topic]" white paper pdf
"[topic]" evaluation report
```

Scholarly discovery expansion

```
site:scholar.google.com "[topic]" "[outcome]"
"[topic]" systematic review
"[topic]" meta-analysis
```

PRISMA-S explicitly recommends reporting online resources, registries, browsing methods, citation searching, full strategies, and deduplication, while PRESS recommends peer-reviewing the search before the main run. [11]

Prioritized sources and evidence collection

For a topic-agnostic review, source quality should be prioritized by **proximity to the underlying evidence** and **institutional authority**. A review should usually prefer primary studies, registered protocols, datasets, patents, standards, regulatory documents, and official statistics over commentary or derivative summaries. That does not mean secondary syntheses are unimportant; they are often essential for orientation and cross-checking, but they should not be the only basis for conclusions. [12]

Source-type comparison

Source type	Best use	Strengths	Typical limitations	Examples
Peer-reviewed journal articles	Empirical findings, theory testing, methods	Usually strongest for original evidence; methods and references are inspectable	Publication lag; selective reporting; topic coverage varies by database	PubMed/MEDLINE for biomedical literature; Scopus and Web of Science for broad citation indexing; IEEE Xplore and ACM DL for engineering/ computing. [13]
Conference proceedings	Fast-moving engineering, computing, applied science	Earlier visibility than journals in some fields	Sometimes lighter peer review or limited detail	IEEE Xplore and ACM DL include conference content. [14]
Official institutional or regulatory sources	Laws, guidance, standards, surveillance, formal recommendations	High authority for current rules, public statistics, and official positions	May be nonbinding guidance rather than law; may not evaluate all evidence equally	FDA guidance search; WHO IRIS; World Bank Open Data. [15]
Registries and repositories	Ongoing studies, preregistration, unpublished or developing evidence	Reduce duplication and help detect missing or unpublished work	Entries may be sponsor-submitted or incomplete	ClinicalTrials.gov, PROSPERO, OSF registrations. [16]
Standards and patents	Technical requirements, prior art, compliance boundaries	Essential for engineering, device, safety, and IP questions	Full standards often paywalled; patents are legal/technical, not proof of effectiveness	ISO search, USPTO Patent Public Search, WIPO PATENTSCOPE. [17]
Reputable industry reports	Market sizing, deployment patterns, vendor landscapes	Useful when public-sector or academic evidence lags practice	Methods may be proprietary; conflicts of interest require caution	Use only when methods, authorship, and sources are transparent. [18]

Source type	Best use	Strengths	Typical limitations	Examples
Search engines and broad discovery tools	Recall expansion, citation chasing, hidden documents	Broad coverage and useful for finding scattered material	Opaque coverage and ranking; indexing updates can take time	Google Scholar includes scholarly materials across fields, languages, and countries, but coverage and updates are not fully transparent. [19]

Topic-based source priority

Topic archetype	Highest-priority sources	Secondary sources
Health or medicine	PubMed/MEDLINE, ClinicalTrials.gov, WHO, regulators, systematic reviews, guidelines	Scopus, Web of Science, preprints, grey literature
Engineering or computing	IEEE Xplore, ACM DL, standards bodies, patents, technical documentation, benchmark papers	Scopus, Web of Science, vendor documentation, reputable industry analyses
Public policy or development	Official government sites, WHO, World Bank, OECD-style datasets, laws, parliamentary or agency reports	Peer-reviewed policy evaluation papers, think tank reports with transparent methods
Legal or regulatory	Statutes, agency guidance, court decisions, standards, official consultation papers	Expert secondary commentary, law reviews
Market or business	Official company filings, regulators, audited reports, trade and customs data, reputable industry datasets	Analyst reports, association reports, news used only for triangulation

Sample data-extraction table

A reusable extraction sheet should collect only fields that support synthesis and decision-making. Cochrane treats data collection as a distinct review stage and links it directly to later synthesis decisions. [20]

Study or source ID	Source type	Jurisdiction or setting	Population or unit of analysis	Intervention, exposure, concept, or policy	Comparator or baseline	Outcome or claim	Key method	Key re
S001	Journal article	US hospitals	Adult patients	Intervention X	Usual care	Mortality, adverse events	RCT	RR 0.8

Study or source ID	Source type	Jurisdiction or setting	Population or unit of analysis	Intervention, exposure, concept, or policy	Comparator or baseline	Outcome or claim	Key method	Key re
S002	Government report	UK	National program	Policy Y	Pre-policy period	Uptake, cost	Administrative analysis	Uptak increa
S003	Patent	Global	Device class	Architecture Z	None	Technical capability claim	Patent filing	Claim novel mech
S004	Qualitative study	Schools	Teachers	Tool A	None	Adoption barriers	Interviews	Traini and workf barrie domir

Synthesis, appraisal, and confidence

After screening and extraction, the synthesis plan should match the question and the evidence actually found. Cochrane distinguishes between preparation for synthesis, meta-analysis, and other synthesis methods; Campbell likewise supports integration of study outcomes, study quality, and evidence mapping, especially in social and policy domains. [21]

A good default synthesis rule is:

- Use **meta-analysis** only when studies are sufficiently similar in design, population, intervention or exposure, and outcomes.
- Use **structured narrative synthesis** when effect estimates cannot be pooled cleanly.
- Use **thematic synthesis** or framework synthesis for qualitative and implementation evidence.
- Use a **mixed-methods convergent synthesis** when quantitative and qualitative evidence answer different parts of the same decision question. [22]

Risk-of-bias appraisal should be matched to design, not applied generically. For randomized trials, RoB 2 assesses bias in five domains and produces “low risk,” “some concerns,” or “high risk” judgments. For non-randomized intervention studies, ROBINS-I assesses seven domains and is designed around the idea of emulating a target trial. For broader quantitative study designs, JBI offers design-specific critical appraisal tools to assess trustworthiness, relevance, and results. [23]

Confidence in the body of evidence should then be assessed separately from individual-study appraisal. GRADE rates certainty across a body of evidence using domains such as risk of bias, inconsistency, indirectness, imprecision, and publication or dissemination bias; randomized studies usually start at high certainty and observational studies usually start at low, though ROBINS-I-based approaches can alter the starting point in some circumstances. For qualitative syntheses, GRADE-CERQual assesses methodological limitations, coherence, adequacy, and relevance. [24]

Sample evidence-confidence matrix

This matrix is an **adapted operational template** based on GRADE for quantitative bodies of evidence and CERQual for qualitative findings. [25]

Confidence level	Typical quantitative signals	Typical qualitative or policy signals	How to write it in the report
High	Low risk of bias, consistent results, direct evidence, precise estimates, low concern for publication bias	Strong methodological transparency, coherent themes across multiple credible sources, high relevance, adequate depth	“The evidence is strong and unlikely to change materially with additional research.”
Moderate	Some study limitations or moderate inconsistency, but core direction is stable	Some limitations in relevance, adequacy, or coherence, but findings still converge	“The evidence supports the conclusion, but future research could change its strength or boundaries.”
Low	Serious bias, indirectness, small samples, or notable inconsistency	Thin or uneven source base, limited methodological transparency, or weak coherence	“The conclusion is plausible but fragile and should be treated as provisional.”
Very low	Critical risk of bias, highly indirect evidence, severe imprecision, or suspected missing evidence	Sparse, contradictory, or poorly documented findings	“The evidence is insufficient for confident decision-making; use only as exploratory context.”

Recommended synthesis outputs

A rigorous report should produce more than prose. At minimum, it should include:

Output	Purpose
Evidence table	Shows what was included and why it matters
Risk-of-bias summary	Separates weak from strong evidence
Confidence table	Distinguishes evidence direction from certainty
Findings synthesis	Answers the predefined key questions
Executive summary	Gives the decision-ready takeaways
Appendices	Preserve search strings, screening logic, exclusions, and extraction rules

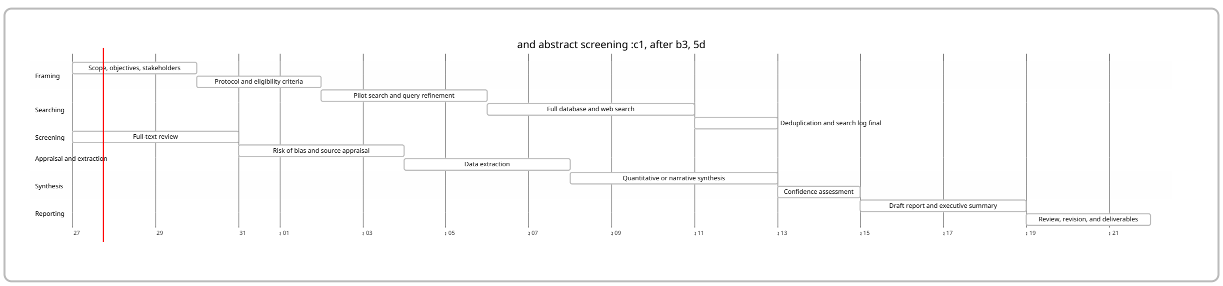
Limitations, timeline, deliverables, and recommended next steps

Even a well-designed deep-research process has recurrent failure modes. The most important are **publication or dissemination bias**, uneven database coverage, language bias, weak grey literature, opaque search-engine indexing, changing regulations, and the possibility that registry content or official guidance may not reflect final or binding reality. GRADE explicitly treats dissemination bias as a certainty-lowering domain; PRISMA-S recommends looking beyond bibliographic databases to registries and web resources; Google Scholar's official guidance makes clear that inclusion and update timing are crawler-dependent; and ClinicalTrials.gov states that records are submitted and maintained by sponsors or investigators, not fully government-authored as evidence summaries. FDA guidance documents, meanwhile, describe the agency's current thinking and are typically nonbinding unless statute or regulation is invoked. [26]

The practical response is to build in safeguards: preregister the scope, peer-review the search, search both databases and official sites, record exclusions, separate empirical evidence from policy or marketing claims, stratify findings by risk of bias, and state confidence levels explicitly. That is what prevents a "deep" report from becoming a polished but unstable summary. [27]

Recommended timeline

The timeline below is a **proposed reusable schedule** for a concise but rigorous deep-research project. It is designed to preserve the key protocol, search, appraisal, and synthesis steps expected by PRISMA-style reporting and Cochrane- or Campbell-style conduct. [28]



Standard deliverables

Deliverable	Recommended content
Protocol memo	Objectives, scope, questions, hypotheses, inclusion/exclusion criteria, planned sources
Search appendix	All search strings exactly as run, dates searched, platforms, limits, deduplication method
Screening log	Inclusion and exclusion decisions at title/abstract and full-text stages
Extraction workbook	Structured source-level data and notes
Evidence appendix	Risk-of-bias judgments, confidence ratings, supporting tables

Deliverable	Recommended content
Main report	Executive summary, findings, synthesis, limitations, and recommendations
Decision brief	One-page version for stakeholders who need action, not methods

Recommended next steps

If a real topic is still unspecified, the best next move is to run a **topic-definition sprint** before doing full review work. The sequence should be:

- Choose the topic and the decision to be supported.
- Decide whether the first pass is a **scoping** review or a **focused systematic** review.
- Write a one-page protocol.
- Pilot one academic query and one official-site query.
- Revise the scope before scaling the search.
- Build extraction and confidence tables before screening begins.

A minimal starter brief can be written in this exact format:

Field	Example placeholder
Topic	"[Specify topic]"
Decision need	"What decision should this research support?"
Primary question	"What do we need to know to make that decision?"
Review type	"Scoping / systematic / rapid / mixed-methods"
Time horizon	"Last 5 years plus foundational studies"
Core databases	"Two multidisciplinary + domain-specific + official sites"
Non-negotiable sources	"Primary studies, official guidance, registries, standards, statistics"
Main outputs	"Executive summary, evidence table, confidence matrix, recommendations"

Used this way, the framework scales from a one-week rapid scan to a multi-week formal review while preserving the same basic logic: **define the question, search transparently, collect systematically, appraise by design, synthesize explicitly, and report confidence separately from conclusions**. [29]

[1] [7] [29] The PRISMA 2020 statement: an updated guideline for reporting systematic reviews | The BMJ
https://www.bmj.com/content/372/bmj.n71.long?utm_source=chatgpt.com

[2] [8] [12] [20] [21] [22] Cochrane Handbook for Systematic Reviews of Interventions | Cochrane Training
https://training.cochrane.org/handbook/current?utm_source=chatgpt.com

[3] [9] [11] PRISMA-S: an extension to the PRISMA Statement for Reporting Literature Searches in Systematic Reviews | Systematic Reviews | Springer Nature Link
https://doi.org/10.1186/s13643-020-01542-z%E2%80%8B?utm_source=chatgpt.com

4 Scoping — PRISMA statement
https://www.prisma-statement.org/scoping?utm_source=chatgpt.com

5 PICO search Help | Cochrane Community
https://community.cochrane.org/pico-search-help?utm_source=chatgpt.com

6 28 Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation | The BMJ
https://doi.org/10.1136/bmj.g7647?utm_source=chatgpt.com

10 PRESS Peer Review of Electronic Search Strategies: 2015 Guideline Statement - ScienceDirect
https://www.sciencedirect.com/science/article/pii/S0895435616000585?utm_source=chatgpt.com

13 About - PubMed
https://pubmed.ncbi.nlm.nih.gov/about/?utm_source=chatgpt.com

14 xplorestaging.ieee.org
https://xplorestaging.ieee.org/Xplorehelp/overview-of-ieee-xplore/about-content?utm_source=chatgpt.com

15 Guidances | FDA
https://www.fda.gov/regulatory-information/search-fda-guidance-documents/guidances?utm_source=chatgpt.com

16 About ClinicalTrials.gov | ClinicalTrials.gov
https://clinicaltrials.gov/about?utm_source=chatgpt.com

17 ISO - Search
https://www.iso.org/cms/%20render/live/en/sites/isoorg/search.html?utm_source=chatgpt.com

18 26 GRADE Book
https://book.gradepro.org/guideline/dissemination-bias?utm_source=chatgpt.com

19 Google Scholar Help
https://scholar.google.co.in/intl/en/scholar/inclusion.html?utm_source=chatgpt.com

23 RoB 2: a revised tool for assessing risk of bias in randomised trials | The BMJ
https://www.bmj.com/content/366/bmj.l4898.abstract?utm_source=chatgpt.com

24 25 Chapter 7: GRADE Criteria Determining Certainty of Evidence | ACIP GRADE Handbook | CDC
https://www.cdc.gov/acip-grade-handbook/hcp/chapter-7-grade-criteria-determining-certainty-of-evidence/index.html?utm_source=chatgpt.com

27 Welcome to Registrations & Preregistrations! - OSF Support
https://help.osf.io/article/330-welcome-to-registrations?utm_source=chatgpt.com